

Kunal Mathur

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GitHub: github.com/kunal202426 | Portfolio: portfolio-kunal26.vercel.app

SUMMARY

Full-Stack Software Engineer (SDE) shipping production features end-to-end on a trading platform handling 20,000+ daily trades at YES Bank. Strong across React, TypeScript, Node.js, Express, and MongoDB, with solid REST API design, microservices, and CI/CD. IIT Madras Global Hyperloop DesignX winner; Flipkart GRiD 7.0 national semifinalist.

EDUCATION

Vellore Institute of Technology

B.Tech, Computer Science & Engineering

2022 - 2026

Vellore, Tamil Nadu, India

EXPERIENCE

YES Securities (YES Bank)

Full Stack Developer Intern

Jan 2026 - Present

Bengaluru, India

- Shipped 5+ production features on the OMNI trading platform (React, Node.js, REST APIs) serving 20,000+ daily trades; refactored backend microservices to cut average API latency ~25%.
- Diagnosed N+1 bottlenecks and redesigned MongoDB schemas to cut query time ~35%; streamlined CI/CD across 3 microservices to speed release cadence.

PROJECTS

CodeGraph: Code Retrieval Engine for AI Agents | GitHub

2026 - Present

- Full-stack retrieval engine modeling a codebase as an entity/edge graph in DuckDB, with a 9-language parsing pipeline (tree-sitter) and a hybrid ranking layer (Reciprocal Rank Fusion) exposed over a 9-tool API; cut client context usage 9.6x.

CityFlow: Real-Time 3D City Analytics Platform | GitHub

Feb 2026 - Apr 2026

- Interactive 3D city simulation (React, Three.js) modeling urban logistics across 5 demand variables, with a live 4-KPI analytics dashboard driven by real-time WebSocket updates on a Node backend.

Hybrid Stock Price Prediction System | GitHub

Jan 2026 - Apr 2026

- End-to-end forecasting app with ETL pipelines ingesting OHLCV market and news data, served via a FastAPI REST layer to a React dashboard; ensemble model (LSTM + LightGBM + FinBERT) cut MAE ~18% vs single-model baselines.

ACHIEVEMENTS

Winner: Global Hyperloop Competition (Asia's First), IIT Madras

2025

- AI & Software Lead, Team Levitate (VIT): built real-time 3D pod simulations (Three.js, React) and hardware telemetry pipelines at sub-50ms refresh; team won the DesignX trophy and 5 category awards among 15+ international teams.

National Semifinalist: Flipkart GRiD 7.0

2025

CERTIFICATIONS

Citi - Software Development - Job Simulation (Forage), 2026

Deloitte - Data Analytics - Job Simulation (Forage), 2025

Data Analytics - GeeksforGeeks, 2026 · 98% score · Python, Pandas, NumPy, SQL, Power BI, Tableau

Blockchain Developer - IBM, 2025 · Smart contracts, Hyperledger Fabric, DApp architecture

McKinsey Forward Program - McKinsey & Company, 2026

SKILLS

Languages: JavaScript, TypeScript, Python, Java, SQL, Data Structures & Algorithms

Frontend: React.js, Next.js, Redux, Three.js, React Flow, HTML, CSS

Backend & APIs: Node.js, Express.js, REST API Design, WebSockets, Microservices, FastAPI, ETL

Databases & DevOps: MongoDB, PostgreSQL, Redis, Firebase, Git, CI/CD, Schema Design

Other: Machine Learning, LLMs, Embeddings, Vector Search, Agile, Figma